STEM Investigate and Play Tray Guidance

What Is STEM?

STEM stands for Science, Technology, Engineering and Mathematics. For young children, these subjects are introduced through hands-on exploration and collaborative experiences, which enable children to develop lifelong thinking skills, such as curiosity and critical thinking. STEM exploration helps to build the foundations of each of these subjects. In early years, this could look like:

- science (observing the natural world, exploring properties of materials, engaging with simple experiments);
- technology (exploring everyday devices, understanding cause and effect);
- engineering (designing and building, problem-solving through construction);
- mathematics (counting and number sense, shape and spatial awareness, measurement).

Why Is STEM Important?

Providing experience of STEM activities enables young children to develop a range of skills, such as:

- communication (asking questions, explaining and talking about their thinking)
- teamwork and collaboration
- creativity
- observation
- problem-solving
- resilience
- making links to previous learning and experience
- working towards a goal

STEM Skills

These STEM Investigate and Play Trays have been designed to encourage younger children to develop a range of skills associated with STEM.

Predict and Plan

Children guess what they think might happen and decide what they need to do and how they will do it. **Example:** Children might explore rolling cars down ramps, predicting what will happen if they change the slope, or planning ways to investigate how to make the cars go faster.

The following prompts may help children to discuss predicting and planning:

- What can you see?
- What do you think will happen if…?
- Can you talk about how you might do that?
- I wonder how you could…

Test and Observe

Children try out their ideas and look closely at what happens. **Example:** When trying to melt ice to free an object, children might think about what they could use to start the melting process and look at the effects.

The following prompts may help children to talk about testing and their observations:

- How can you test…?
- What are you testing?
- Why is that helpful?
- What do you notice?
- What did you find out?
- I wonder how we will know if…

Analyse and Evaluate

Children think about whether their ideas are working and if they need to make any changes. **Example:** When building structures, children might decide that some shapes or materials aren't suitable for their intended purpose and opt to try different pieces next time.

Children could be encouraged to analyse and evaluate their experience with the following prompts:

- Are you happy with what happened?
- How well did this work?
- Is there anything you would do differently next time?
- What worked best and why?
- Did anything unexpected happen?

Throughout all of the skills above, children can **make links** to their previous experiences by connecting what they already know to new situations, helping them understand problems more deeply and build on their existing learning.